



Caringbah High School

Year 12 Mathematics Extension 1 Assessment Task 2 Informal Examination 2021

General Instructions

- Reading time – 5 minutes
- Working time – 75 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- A reference sheet is provided separately
- In Questions 6-9, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 60

Section I 5 marks

Attempt Questions 1-5

Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 55 marks

Attempt Questions 6-9

Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Section I					Total	
Q 1-5	Q6	Q7	Q8	Q9		
/5	/13	/14	/14	/14	/60	%

Section I -5 marks

Attempt Questions 1-5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5

1 If $\underline{a} = -7\underline{i} + 2\underline{j}$, what does $|2\underline{a}|$ equal?

A) $2\sqrt{53}$

B) $3\sqrt{41}$

C) $-14\underline{i} + 4\underline{j}$

D) $14\underline{i} - 4\underline{j}$

2 If $y = \tan^{-1}(\sin x)$, then $\frac{dy}{dx} = ?$

A) $\frac{1}{1 + \sin^2 x}$

B) $\sec^2 x$

C) $\frac{\cos x}{1 + \sin^2 x}$

D) $\sec x$

3 Using the substitution $u = \ln x$ or otherwise, which of the following is the primitive of $\frac{(\ln x)^2}{x}$?

A) $\frac{x(\ln x)^3}{3} + C$

B) $\frac{(\ln x)^3}{3x} + C$

C) $(\ln x)^3 + C$

D) $\frac{(\ln x)^3}{3} + C$

4 Consider the vector $\vec{a} = \begin{bmatrix} 2x-1 \\ 1-2x \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 2x+1 \\ 1+2x \end{bmatrix}$. Are these vectors $\vec{a}$ and $\vec{b}$ perpendicular?

A) No

B) Yes

C) Not enough information

D) Vectors are parallel

5 What is the integral $\int_1^2 x\sqrt{2-x} \, dx$ transformed to by the substitution $u = \sqrt{2-x}$?

A) $\int_1^2 (2u - u^3) \, du$

B) $\int_1^0 (2u - u^3) \, du$

C) $\int_1^2 (2u^4 - 4u^2) \, du$

D) $\int_1^0 (2u^4 - 4u^2) \, du$

End of Section I

Section II

55 marks

Attempt Questions 6–9

Allow about 67 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (13 Marks)

Marks

a) Differentiate

i) $y = \tan^{-1} x^3$ 1

ii) $y = 2x^2 \sin^{-1}(2x-1)$ 2

iii) $y = \cos^{-1}(\log_e x)$ 1

b) Let $\underline{u} = 4\underline{i} - 2\underline{j}$ and $\underline{v} = 2\underline{i} - 5\underline{j}$, find

i) $\underline{u} - \underline{v}$ 1

ii) $\underline{u} \cdot \underline{v}$ 1

iii) $\hat{\underline{u}}$ 1

c) The curve $y = \frac{1}{\sqrt{x^2+9}}$ is rotated about the x -axis. Find the volume of the solid of revolution formed between $x=0$ and $x=3\sqrt{3}$. 3

d) Find λ and μ such that $\lambda \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \mu \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. 3

End of Question 6

Question 7 (14 Marks)

a) If $f(x) = \sin^{-1} x + \cos^{-1} x$ and $-1 \leq x \leq 1$,

i) By using $f'(x)$ show that $f(x) = \frac{\pi}{2}$ for all x in the given domain.

2

ii) Hence evaluate $\int_{-1}^1 f(x) dx$

1

b) Find the vector projection of $\underline{u} = 2\hat{i} - 5\hat{j}$ onto $\underline{v} = \hat{i} - \hat{j}$

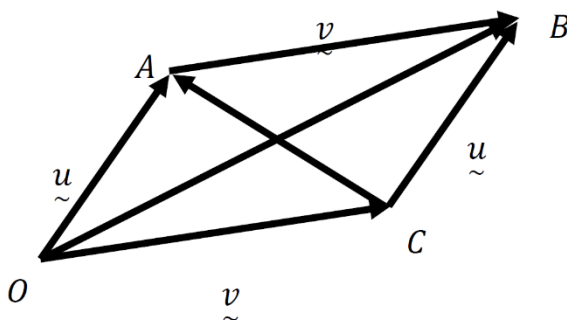
2

c) Evaluate $\int_0^{\frac{1}{\sqrt{2}}} \sqrt{1-x^2} dx$ using the substitution $x = \sin u$

3

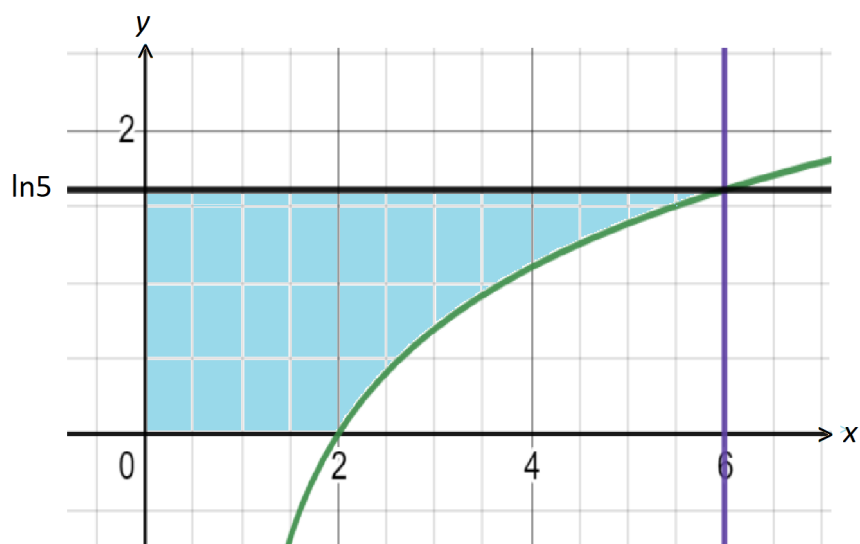
d) In the parallelogram $OABC$, $\overrightarrow{OA} = \overrightarrow{CB} = \underline{u}$ and $\overrightarrow{OC} = \overrightarrow{AB} = \underline{v}$. Prove that the diagonals of a parallelogram intersect at right angles if and only if it is a rhombus.

2



Question 7 continued on page 6

- e) In the diagram, the shaded region is bounded by $y = \log_e(x-1)$, the coordinate axes and the line $y = \ln 5$. This region is then rotated about the y -axis.



- i) Show that the volume of the solid of revolution of the shaded region formed is given by 2

$$V = \pi \int_0^{\log_e 5} (1 + 2e^y + e^{2y}) dy$$

- ii) Evaluate, in exact form, the volume of the solid. 2

End of Question 7

Question 8 (14 Marks)

a) Find

i) $\int \frac{e^{2x}}{\sqrt{64 - e^{4x}}} dx$ 2

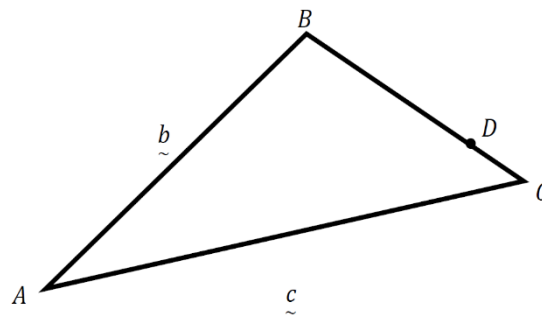
ii) $\int \frac{dx}{\sqrt{4 - 9x^2}}$ 2

b) Find the angle, to the nearest degree, between $\underline{u} = 2\underline{i} - 3\underline{j}$ and $\underline{v} = 2\underline{i} - \underline{j}$ 2

c) i) Show that $x^2 + 2x + 5 = (x + 1)^2 + 4$ 1

ii) Hence find $\int_{-1}^1 \frac{1}{x^2 + 2x + 5} dx$ 2

d) In the diagram, $\overrightarrow{AB} = \underline{b}$ and $\overrightarrow{AC} = \underline{c}$. D divides BC in the ratio 5:1. Find an expression for $\overrightarrow{AD}$ in terms of $\underline{b}$ and $\underline{c}$. 3



e) Find the vector $\underline{v}$ parallel to $\underline{u} = -2\underline{i} + 5\underline{j}$ that has a magnitude of 4. 2

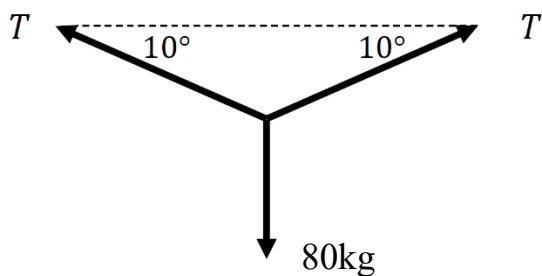
End of Question 8

Question 9 (14 Marks)**Marks**

a) Evaluate $\int_0^{\frac{\pi}{2}} \sin^3 x \cos x \, dx$

2

- b) A hammock of negligible mass is attached to two trees, each at an angle of 10° to the horizontal, with the tension in the ropes at both sides equal. A person of mass 80 kg is lying in the hammock. What is the tension T , assuming $g = 9.8 \text{ ms}^{-2}$.

3

- c) The gradient at the point (x, y) of the function $y = f(x)$ is given by $\frac{dy}{dx} = \sin^2 x$. The curve passes through the point $(0, -2)$. Find y when $x = \frac{\pi}{4}$.

3

d) Evaluate $\int_0^1 \frac{(6\sqrt{x}-1)^2}{\sqrt{x}} \, dx$, using the substitution $u = 6\sqrt{x}-1$

3

e) If $y = x \cos^{-1} 2x - \frac{1}{2} \sqrt{1-4x^2}$, find $\frac{dy}{dx}$, simplified fully.

3

End of Question 9
END OF EXAMINATION!

Name: _____

Mathematics Extension 1

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct
 ↗

-
- Start ➔ 1. A ☐ B ☐ C ☐ D ☐
Here
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐

Yr 12 Mathematics Ext 1 AT2 2021

Mult. Choice

Q1 $\underline{a} = -7\underline{i} + 2\underline{j}$

$$|\underline{a}| = \sqrt{(-7)^2 + 2^2}$$

$$= \sqrt{53}$$

$\therefore |\underline{2a}| = 2\sqrt{53}$ (A)

Q2 $y = \tan^{-1}(\sin x)$

$$\frac{dy}{dx} = \frac{\cos x}{1 + \sin^2 x} \quad \text{(C)}$$

Q3 $\int \frac{(\ln x)^2}{x} dx$

$$= \int u^2 du$$

$$= \frac{u^3}{3} + c$$

$$= \frac{(\ln x)^3}{3} + c \quad \text{(D)}$$

Q4 $\underline{a} = \begin{bmatrix} 2x-1 \\ 1-2x \end{bmatrix}, \underline{b} = \begin{bmatrix} 2x+1 \\ 1+2x \end{bmatrix}$

$$\underline{a} \cdot \underline{b} = (2x-1)(2x+1) + (1-2x)(1+2x)$$

$$= 4x^2 - 1 + 1 - 4x^2$$

$$= 0$$

$\therefore \underline{a} \perp \underline{b}$ (B)

Q5 $\int_1^2 x\sqrt{2-x} dx$

$$u = \sqrt{2-x}$$

$$u^2 = 2-x$$

$$x = 2-u^2$$

$$= \int_1^0 (2-u^2)(u)(-2u) du$$

$$\frac{dx}{du} = -2u$$

$$= \int_1^0 (2-u^2)(-2u^2) du$$

$$dx = -2u du$$

$$= \int_1^0 (-4u^2 + 2u^4) du$$

when $x=2, u=0$

" $x=1, u=1$

$$= \int_1^0 (2u^4 - 4u^2) du \quad \text{(D)}$$

Q6

(a)(i) $y = \tan^{-1} x^3$

$$\frac{dy}{dx} = \frac{3x^2}{1+(x^3)^2}$$

$$= \frac{3x^2}{1+x^6} \quad \text{(1)}$$

(ii) $y = 2x^2 \sin^{-1}(2x-1)$

$$\frac{dy}{dx} = 2x^2 \times \frac{2}{\sqrt{1-(2x-1)^2}} + \sin^{-1}(2x-1)(4x)$$

$$= \frac{4x^2}{\sqrt{1-(2x-1)^2}} + 4x \sin^{-1}(2x-1)$$

$$= \frac{4x}{\sqrt{4x-4x^2}} + 4x \sin^{-1}(2x-1)$$

$$= \frac{2x}{\sqrt{x-x^2}} + 4x \sin^{-1}(2x-1)$$

(iii) $y = \cos^{-1}(\log_e x)$

$$\frac{dy}{dx} = -\frac{\frac{1}{x}}{\sqrt{1-(\log_e x)^2}}$$

$$= -\frac{1}{x\sqrt{1-(\log_e x)^2}} \quad \text{(1)}$$

Q6 cont

$$(b) \underline{u} = 4\underline{i} - 2\underline{j}, \underline{v} = 2\underline{i} - 5\underline{j}$$

$$(i) \underline{u} - \underline{v} = (4\underline{i} - 2\underline{j}) - (2\underline{i} - 5\underline{j})$$

$$= 2\underline{i} + 3\underline{j} \quad (1)$$

$$(ii) \underline{u} \cdot \underline{v} = 4(2) + (-2)(-5)$$

$$= 8 + 10$$

$$= 18 \quad (1)$$

$$(iii) \hat{u} = \frac{4\underline{i} - 2\underline{j}}{\sqrt{4^2 + (-2)^2}}$$

$$= \frac{4\underline{i} - 2\underline{j}}{\sqrt{16 + 4}}$$

$$= \frac{4\underline{i} - 2\underline{j}}{\sqrt{20}} \Rightarrow \hat{u} = \frac{2}{\sqrt{5}}\underline{i} - \frac{1}{\sqrt{5}}\underline{j} \quad (1)$$

$$= 2\sqrt{5}$$

(c) $y = \frac{1}{\sqrt{x^2 + 9}}$, rotated about
x-axis, $x=0$, $x=3\sqrt{3}$.

$$V = \pi \int_0^{3\sqrt{3}} \frac{1}{x^2 + 9} dx \quad y^2 = \frac{1}{x^2 + 9} \quad (1)$$

$$= \pi \left[\frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) \right]_0^{3\sqrt{3}} \quad (1)$$

$$= \frac{\pi}{3} \left[\tan^{-1}\left(\frac{3\sqrt{3}}{3}\right) - \tan^{-1}(0) \right]$$

$$= \frac{\pi}{3} \left[\tan^{-1}(\sqrt{3}) - \tan^{-1}(0) \right]$$

$$= \frac{\pi}{3} \left[\frac{\pi}{3} - 0 \right]$$

$$= \frac{\pi^2}{9} \quad (1)$$

$$(d) \lambda \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \mu \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$2\lambda + 4\mu = 0 \quad (1) \times 3$$

$$3\lambda + 5\mu = 1 \quad (2) \times 2$$

$$6\lambda + 12\mu = 0$$

$$6\lambda + 10\mu = 2$$

$$2\mu = -2$$

$$\mu = -1 \quad (1)$$

sub $\mu = -1$ into (1)

$$2\lambda - 4 = 0$$

$$2\lambda = 4$$

$$\lambda = 2 \quad (1)$$

$$\therefore \lambda = 2, \mu = -1$$

Q7

(a) $f(x) = \sin^{-1}x + \cos^{-1}x, -1 \leq x \leq 1$

(i) $f'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}}$

$$= 0 \quad (1)$$

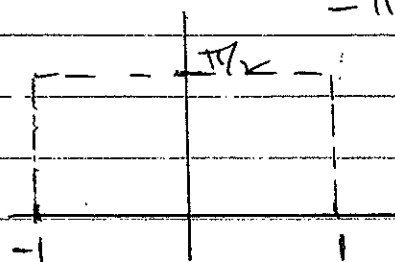
Let $x=0$, $f(0) = \sin^{-1}0 + \cos^{-1}0$

$$= \frac{\pi}{2} \quad (1)$$

$\therefore f(x) = \frac{\pi}{2}$ for all $x, -1 \leq x \leq 1$

(ii) $\int_{-1}^1 f(x) dx = 2 \times \frac{\pi}{2}$

$$= \pi$$



$$(1)$$

Q7 cont.

$$(b) \underline{u} = 2\underline{i} - 5\underline{j}, \underline{v} = \underline{i} - \underline{j}$$

$$\text{proj}_{\underline{v}} \underline{u} = \frac{\underline{u} \cdot \underline{v}}{\underline{v} \cdot \underline{v}} \underline{v}$$

$$= \frac{2(1) + (-5)(-1)}{1^2 + (-1)^2} (\underline{i} - \underline{j}) \quad (1)$$

$$= \frac{7}{2} (\underline{i} - \underline{j})$$

$$= \frac{7}{2} \underline{i} - \frac{7}{2} \underline{j} \quad (1)$$

$$(c) \int_0^{\frac{1}{\sqrt{2}}} \sqrt{1-x^2} dx \quad x = \sin u$$

$$= \int_0^{\frac{\pi}{4}} \sqrt{1-\sin^2 u} x \cos u du \quad dx = \cos u du \quad (1)$$

$$= \int_0^{\frac{\pi}{4}} \sqrt{\cos^2 u} x \cos u du \quad \text{when } x = \frac{1}{\sqrt{2}}, u = \frac{\pi}{4}$$

$$= \int_0^{\frac{\pi}{4}} (\cos u x \cos u) du \quad " \quad x=0, u=0$$

$$= \int_0^{\frac{\pi}{4}} \cos^2 u du \quad (1)$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{2} + \frac{1}{2} \cos 2u du$$

$$= \left[\frac{1}{2} u + \frac{1}{4} \sin 2u \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} \left(\frac{\pi}{4} \right) + \frac{1}{4} \sin \frac{\pi}{2} - 0$$

$$= \frac{\pi}{8} + \frac{1}{4} (1)$$

$$= \frac{\pi}{8} + \frac{1}{4} = \frac{\pi+2}{8} \quad (1)$$

$$(d) \vec{OB} = \underline{u} + \underline{v} \neq \vec{CA} = \underline{u} - \underline{v}$$

If diagonals are $\perp$, $\underline{u} \cdot \underline{v} = 0$

$$(\underline{u} + \underline{v}) \cdot (\underline{u} - \underline{v}) = 0 \quad (1)$$

$$\underline{u}^2 - \underline{v}^2 = 0$$

$$|\underline{u}| - |\underline{v}| = 0$$

$$|\underline{u}| = |\underline{v}|$$

$$\therefore \underline{u} = \underline{v} \quad (1)$$

$\therefore$ the sides are $=$ & the parallelogram must be a rhombus if the diagonals meet at 90° .

$$(e) V = \pi \int_a^b x^2 dy \quad y = \log_e(x+1)$$

$$(i) V = \pi \int_0^{\ln 5} (e^{2y} + 2e^y + 1) dy \quad \begin{cases} x-1 = e^y \\ x = e^y + 1 \\ x^2 = (e^y + 1)^2 \end{cases} \quad (1)$$

$$(ii) V = \pi \left[\frac{1}{2} e^{2y} + 2e^y + y \right]_0^{\ln 5} \quad (1)$$

$$= \pi \left[\left(\frac{1}{2} e^{2 \ln 5} + 2e^{\ln 5} + \ln 5 \right) - \left(\frac{1}{2} e^0 + 2e^0 + 0 \right) \right]$$

$$\left(\frac{1}{2} e^0 + 2e^0 + 0 \right)$$

$$= \pi \left[\frac{1}{2} (25) + 2(5) + \ln 5 - \frac{1}{2} - 2 \right]$$

$$= \pi \left[\frac{25}{2} + 10 + \ln 5 - \frac{1}{2} - 2 \right]$$

$$= \pi (20 + \ln 5) u^3 \quad (1)$$

Q8

$$(a)(i) \int \frac{e^{2x}}{\sqrt{64 - e^{4x}}} dx$$

$$= \frac{1}{2} \int \frac{2e^{2x}}{\sqrt{8^2 - (e^{2x})^2}} dx \quad (1)$$

$$= \frac{1}{2} \sin^{-1} \left(\frac{e^{2x}}{8} \right) + C \quad (1)$$

$$(ii) \int \frac{dx}{\sqrt{4 - 9x^2}}$$

$$= \frac{1}{3} \int \frac{dx}{\sqrt{2^2 - 3^2 x^2}} \quad (1)$$

$$= \frac{1}{3} \sin^{-1} \left(\frac{3x}{2} \right) + C \quad (1)$$

$$(b) \vec{u} = 2\hat{i} - 3\hat{j}, \vec{v} = 2\hat{i} - \hat{j}$$

$$\cos \theta = \frac{2(2) + (-3)(-1)}{\sqrt{2^2 + (-3)^2} \times \sqrt{2^2 + (-1)^2}} \quad (1)$$

$$= \frac{4+3}{\sqrt{13} \times \sqrt{5}}$$

$$= \frac{7}{\sqrt{65}}$$

$$\therefore \theta = \cos^{-1} \left(\frac{7}{\sqrt{65}} \right)$$

$$= 30^\circ$$

(1)

$$(c)(i) x^2 + 2x + 5 = (x+1)^2 + 4$$

LHS:

$$x^2 + 2x + 5 = x^2 + 2x + 1 + 4$$

$$= (x+1)^2 + 4$$

= RHS.

$$\therefore \text{LHS} = \text{RHS}$$

(1)

$$(c)(ii) \int_{-1}^1 \frac{1}{x^2 + 2x + 5} dx$$

$$= \int_{-1}^1 \frac{1}{(x+1)^2 + 4} dx$$

$$= \int_{-1}^1 \frac{1}{4 + (x+1)^2} dx$$

$$= \frac{1}{2} \left[\tan^{-1} \left(\frac{x+1}{2} \right) \right]_{-1}^1 \quad (1)$$

$$= \frac{1}{2} \left[\tan^{-1} \left(\frac{2}{2} \right) - \tan^{-1} (0) \right]$$

$$= \frac{1}{2} \left[\tan^{-1} (1) - \tan^{-1} (0) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{4} - 0 \right]$$

$$= \frac{\pi}{8}$$

(1)

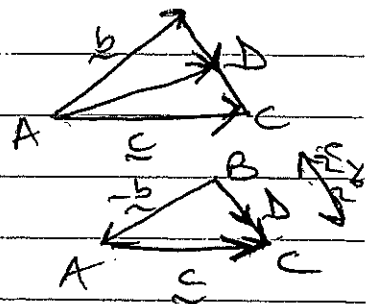
$$(d) \vec{AD} = \vec{AB} + \vec{BD}$$

$$= \vec{AB} + \frac{5}{6} \vec{BC} \quad (1)$$

$$= \vec{b} + \frac{5}{6} (\vec{c} - \vec{b}) \quad (1)$$

$$= \vec{b} + \frac{5}{6} \vec{c} - \frac{5}{6} \vec{b}$$

$$= \frac{1}{6} \vec{b} + \frac{5}{6} \vec{c} \quad (1)$$



(e) $\underline{u} = -2\underline{i} + 5\underline{j}$, magnitude = 4

$$|\underline{u}| = \sqrt{(-2)^2 + 5^2}$$

$$= \sqrt{29}$$

$$\underline{\hat{u}} = \frac{-2}{\sqrt{29}}\underline{i} + \frac{5}{\sqrt{29}}\underline{j} \quad (1)$$

$$\therefore \underline{v} = \underline{\hat{u}} \times (\pm 4)$$

$$\underline{v} = \underline{\hat{u}} \times 4$$

$$= 4 \left(\frac{-2}{\sqrt{29}}\underline{i} + \frac{5}{\sqrt{29}}\underline{j} \right)$$

$$= \frac{-8}{\sqrt{29}}\underline{i} + \frac{20}{\sqrt{29}}\underline{j}$$

OR

$$\underline{v} = \underline{\hat{u}} \times (-4) \quad (1)$$

$$= -4 \left(\frac{-2}{\sqrt{29}}\underline{i} + \frac{5}{\sqrt{29}}\underline{j} \right)$$

$$= \frac{8}{\sqrt{29}}\underline{i} - \frac{20}{\sqrt{29}}\underline{j}$$

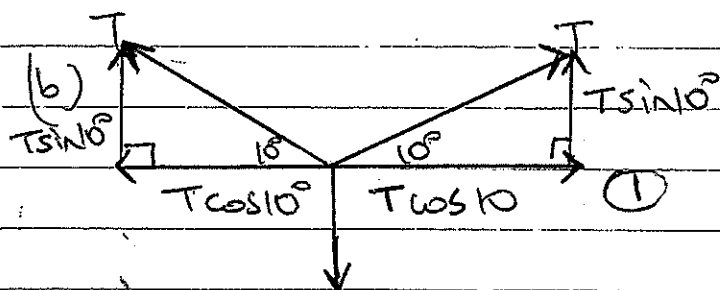
$$\int_0^{\pi/2} \sin^3 x \cos x dx$$

$$= \left[\frac{\sin^4 x}{4} \right]_0^{\pi/2} \quad (1)$$

$$= \frac{1}{4} \left[\sin^4 \left(\frac{\pi}{2} \right) - \sin^4(0) \right]$$

$$= \frac{1}{4} (1 - 0)$$

$$= \frac{1}{4} \quad (1)$$



$$2T \sin 10^\circ = 80 \times 9.8$$

$$T = \frac{40 \times 9.8}{\sin 10^\circ} \quad (1)$$

$$T = 2257 \text{ N} \quad (1)$$

(c) $\frac{dy}{dx} = \sin^2 x$, (0, 2)

when $x = \frac{\pi}{4}$, $y = ?$

$$y = \int \sin^2 x dx$$

$$y = \int \left(\frac{1}{2} - \frac{1}{2} \cos 2x \right) dx \quad (1)$$

$$= \frac{1}{2}x - \frac{1}{4} \sin 2x + C$$

$$-2 = 0 - \frac{1}{4} \sin 0 + C$$

$$C = -2$$

Q9(c) cont.

$$\therefore y = \frac{1}{2}x - \frac{1}{4}\sin 2x - 2 \quad (1)$$

when $x = \frac{\pi}{4}$,

$$y = \frac{1}{2}\left(\frac{\pi}{4}\right) - \frac{1}{4}\sin\frac{\pi}{2} - 2$$

$$y = \frac{\pi}{8} - \frac{1}{4} - 2$$

$$= \frac{\pi}{8} - \frac{9}{4}$$

$$y = \frac{\pi - 18}{8} \quad (1)$$

(d) $\int_0^1 \frac{(6\sqrt{x}-1)^2}{\sqrt{x}} dx$, $u = 6\sqrt{x}-1$

$$\frac{du}{dx} = \frac{3}{\sqrt{x}} \quad (1)$$

$$\frac{1}{3} \int_0^1 \frac{3(6\sqrt{x}-1)^2}{\sqrt{x}} dx \quad du = \frac{3}{\sqrt{x}} dx$$

$$= \frac{1}{3} \int_1^5 u^2 du \quad (1) \quad \begin{array}{l} \text{when } x=1, u=5 \\ \text{" } x=0, u=1 \end{array}$$

$$= \frac{1}{3} \left[\frac{u^3}{3} \right]_1^5$$

$$= \frac{1}{9} (5^3 - 1^3)$$

$$= \frac{1}{9} (126)$$

$$= 14 \quad (1)$$

(e) $y = x \cos^{-1} 2x - \frac{1}{2} \sqrt{1-4x^2}$

$$y = x \cos^{-1} 2x - \frac{1}{2} (1-4x^2)^{1/2}$$

$$\frac{dy}{dx} = x \left(\frac{-2}{\sqrt{1-4x^2}} \right) + \cos^{-1}(2x) (1) \quad (1)$$

$$- \frac{1}{4} (1-4x^2)^{-1/2} (-8x)$$

$$= -\frac{2x}{\sqrt{1-4x^2}} + \cos^{-1}(2x) + \frac{2x}{\sqrt{1-4x^2}} \quad (1)$$

$$= \cos^{-1}(2x) \quad (1)$$